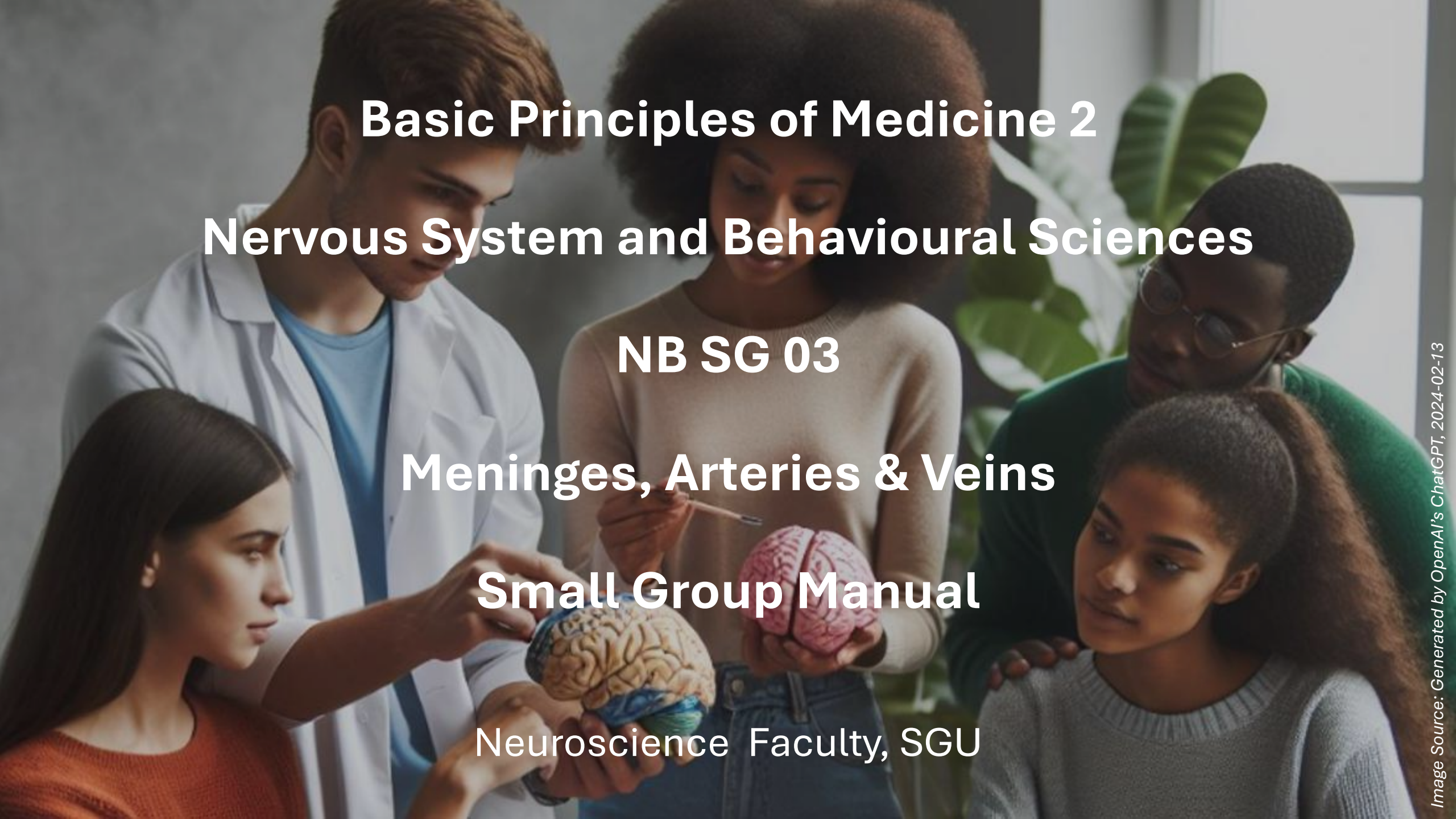




Basic Principles of Medicine 2

Nervous System and Behavioural Sciences

NB SG 03

Meninges, Arteries & Veins

Small Group Manual

Neuroscience Faculty, SGU

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Learning Objectives

By the end of this small group session a student should be able to:

SOM.MK.I.BPM2.5.NB.5.NSCI.0354 Examine the site of production and flow of cerebrospinal fluid on anatomical models of the brain

SOM.MK.I.BPM2.5.NB.5.NSCI.0355 Associate the pituitary gland, pineal gland and area postrema on the anatomical model of a mid-sagittal section of the head with their functional aspects

SOM.MK.I.BPM2.5.NB.5.NSCI.0356 Examine the structural organization of the meninges

SOM.MK.I.BPM2.5.NB.5.NSCI.0357 Examine the structural organization of the subarachnoid space

SOM.MK.I.BPM2.5.NB.5.NSCI.0358 Appraise the external features of the intracranial dura

SOM.MK.I.BPM2.5.NB.5.NSCI.0359 Differentiate the intracranial dura and spinal cord dura

SOM.MK.I.BPM2.5.NB.5.NSCI.0360 Examine the subarachnoid arteries

SOM.MK.I.BPM2.5.NB.5.NSCI.0361 Examine the structural organization of the system of veins and venous sinuses associated with the brain

Relevant Reading

Associated lectures:

- NB 04 – Meninges and arteries
- NB 06 – Transport within the CNS

Neuroanatomy Atlas in Clinical Context (10th edition) Duane E. Haines

- Chapter 2: External morphology of the central nervous system
- Chapter 4: Meninges, cisterns, ventricles, and related hemorrhages

Essential Neuroscience (4th edition) Allan Siegel, Hreday Sapru

- Chapter 3: Meninges and Cerebrospinal Fluid
- Chapter 26: Blood Supply of the Central Nervous System

What to bring to the Small Group

- Electronic devices, such as laptops or tablet computers, with access to:
 - Small group manuals,
 - Related lecture handouts,
 - Neuroanatomy atlas,
 - Neuroanatomy textbook

What are the Expectations

- Before the small group session
 - Review NB lectures 2, 3, and 4
- During the small group session *)
 - Challenge yourself to work together with your peers through the material
 - Listen to your peers, ask questions, and collaboratively drive the discussion
 - Do not expect the facilitators to provide a lecture/review of the content
 - Do not expect the facilitators to drive the discussion without the group's input
 - Complete the facilitator evaluation at the end of the session

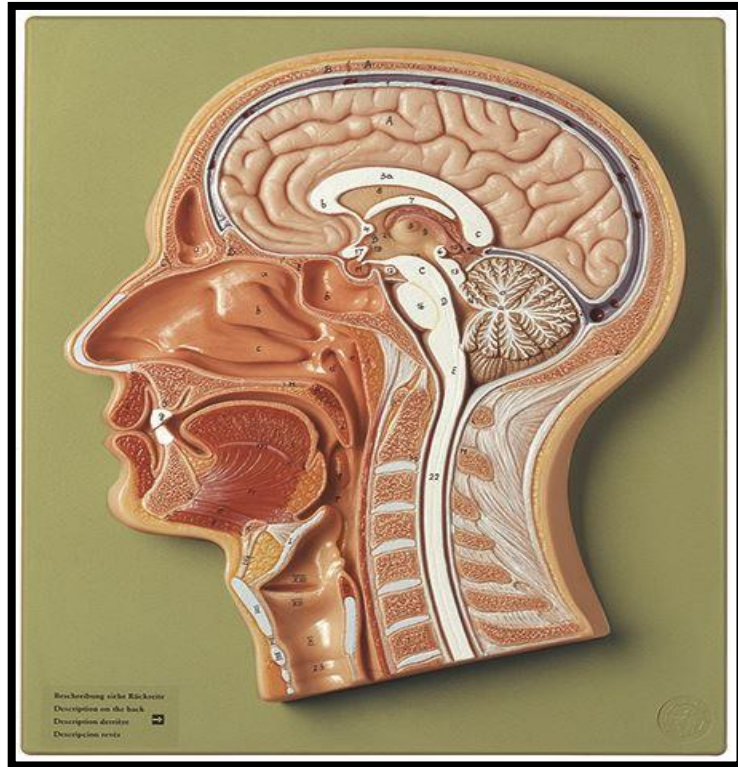
*) From: Learning Pathway (MyCourses BPM2 > Resources > General Information > Your Learning Pathway)

Sequence of Small Group Activities

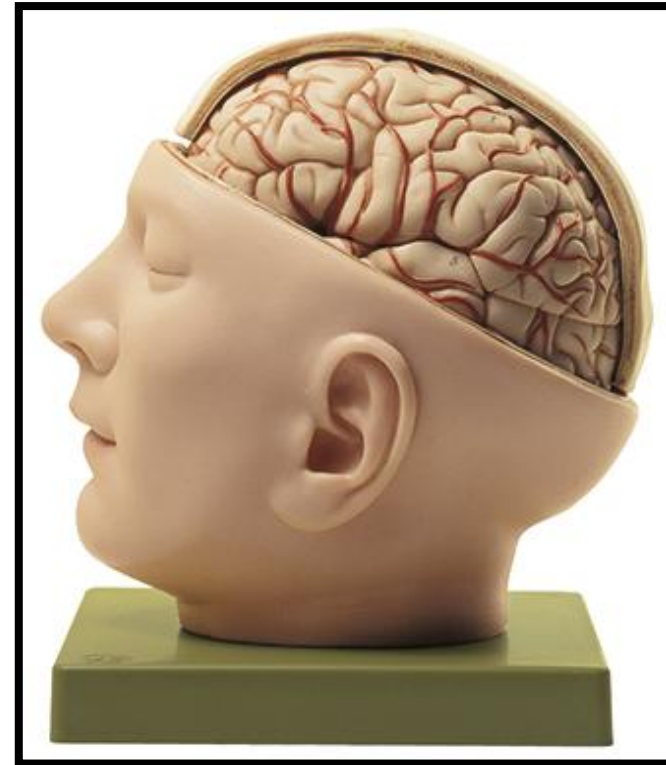
1. Students split into two subgroups of approximately 4 students each and work through the questions using the models and Haines atlas.
2. Individual students present as many structures as they can identify to their subgroup, including the functions of certain structures:
3. The Clinical Instructor presents the Clinical Case Video on the screen
4. In an interactive session, students answer the case-related questions under the guidance of the Clinical Instructor

Please note: The role of the Clinical Instructor is to guide students through the material by asking questions, or breaking down the problems into smaller components, but the focus is on student activity and participation. Come prepared, and don't expect another lecture by the Instructor.

Anatomical Models provided in Small Group



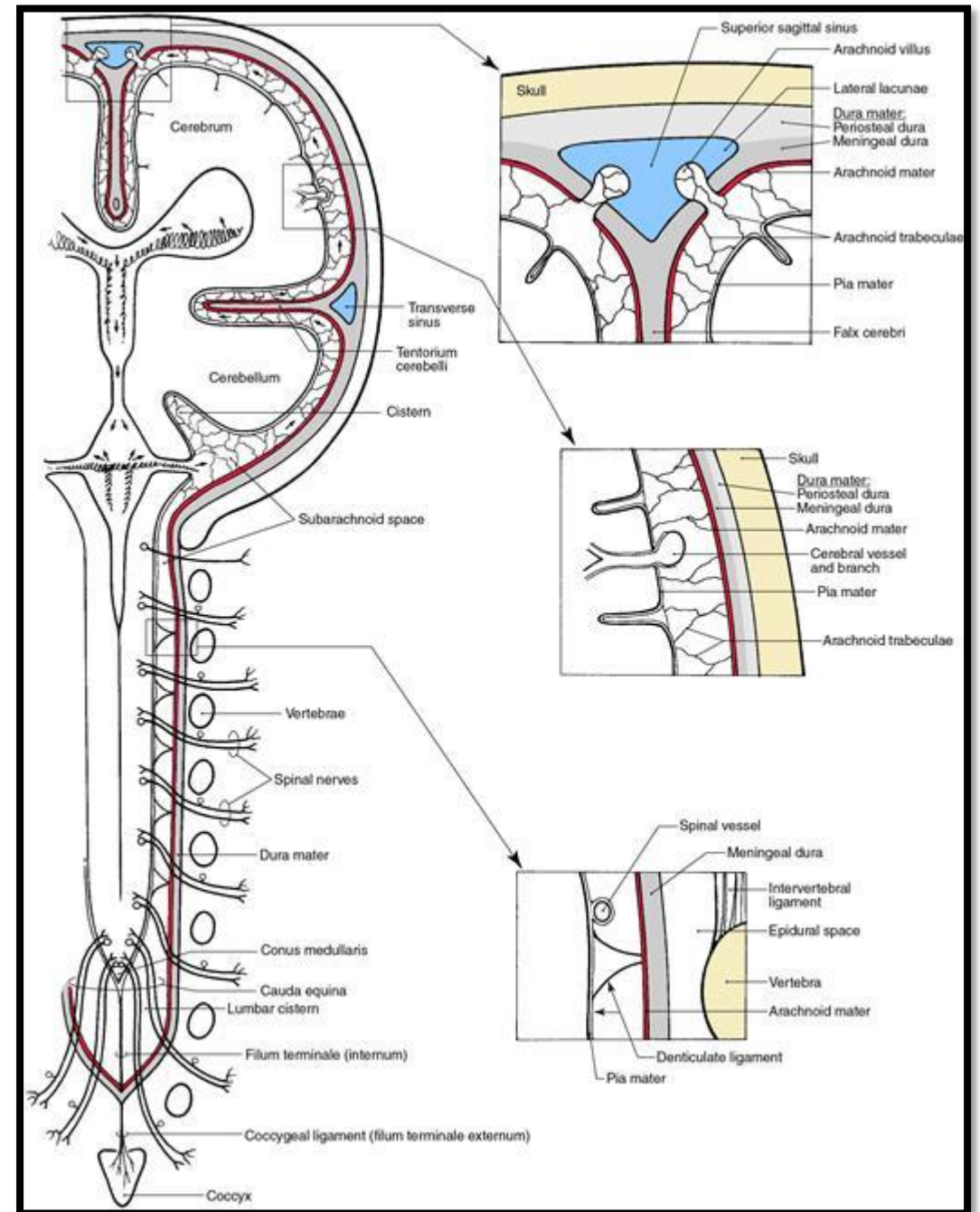
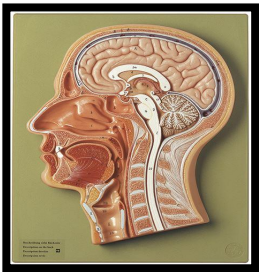
Model: Median Section of the Head



Model: Base of Head with Brain, Falx

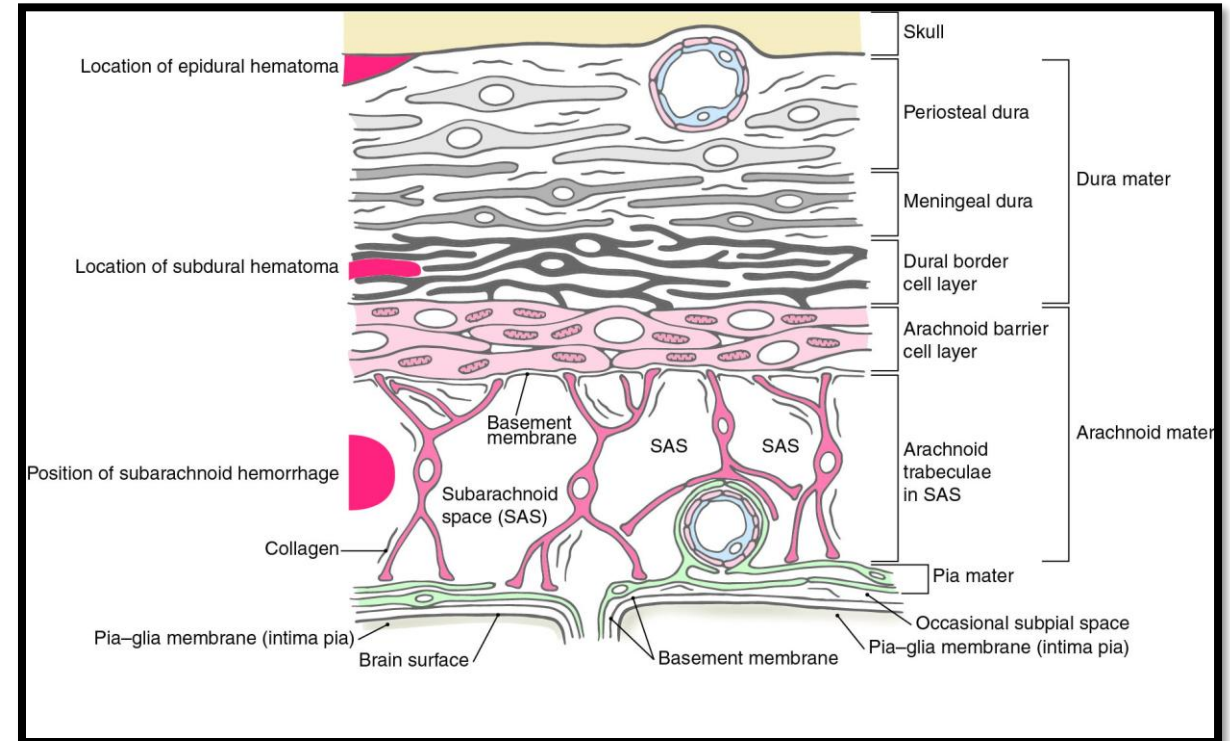
Meninges

- After a street brawl, a young man sustained a knife wound to the head that penetrated the brain tissue. With reference to the image, name the layers from outside to inside that the knife penetrated.
- Locate these layers on the model provided.
- Identify the subarachnoid space and the major structures in that space.

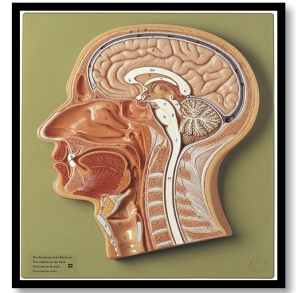


Meninges

- Identify on the image, the location of the four types of intracranial hemorrhages.
- With reference to the image, between which layers would the basilar and vertebral arteries be found?



Circumventricular Organs

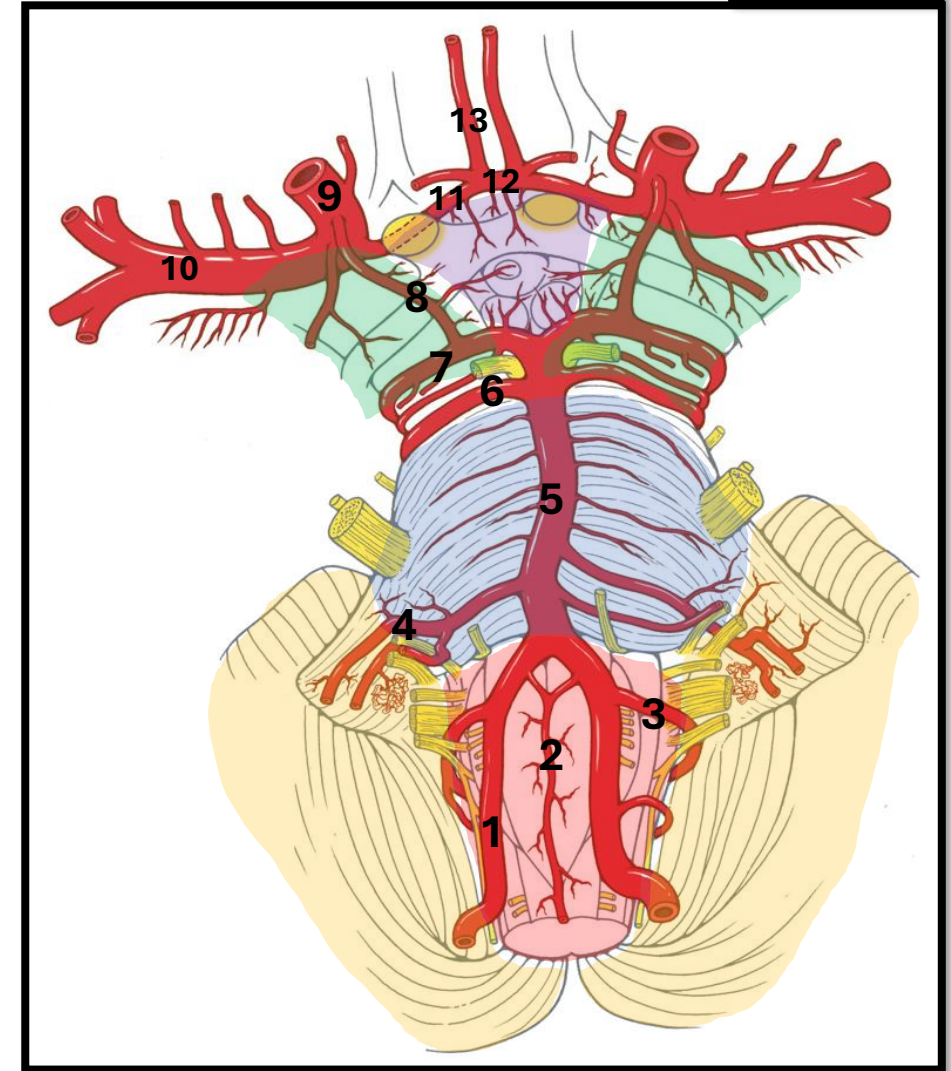
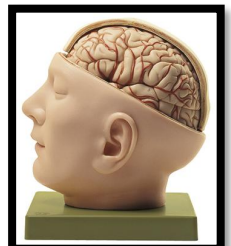


- Identify the locations of the following circumventricular organs:
 - Area Postrema
 - Pineal Gland
 - Pituitary Gland
- What do these structures have in common?
- What happens when these organs lose their function?

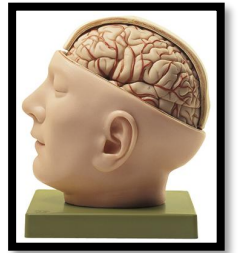
Arteries

Use the atlas and models to identify the numbered arteries

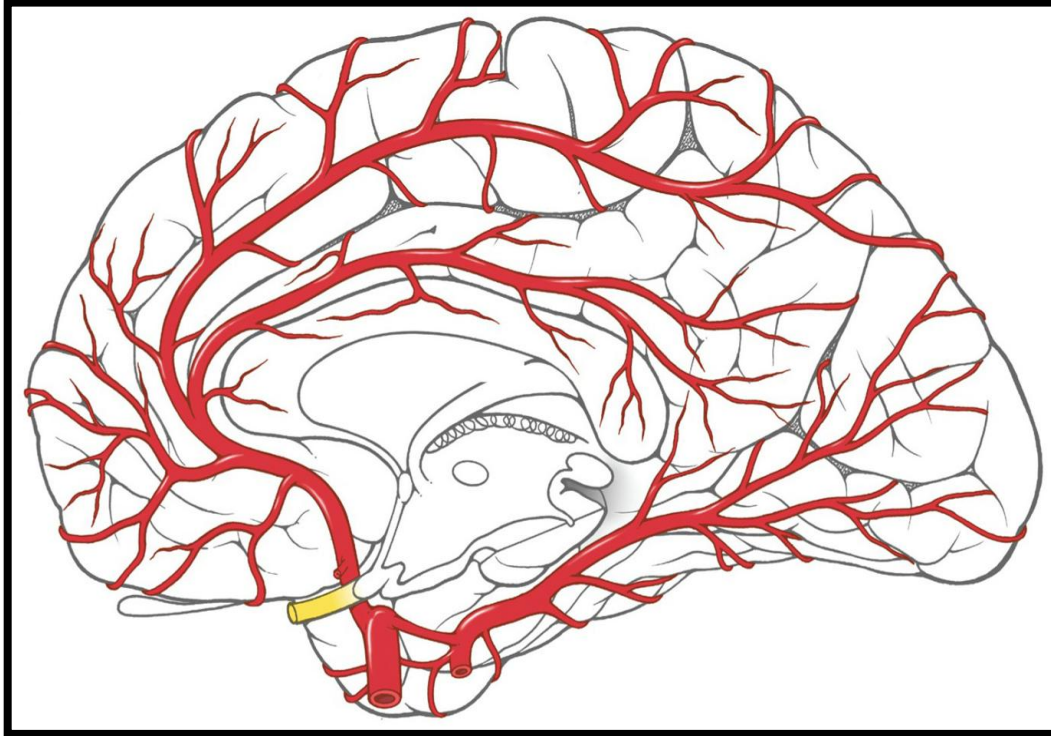
#	Name of Artery
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	



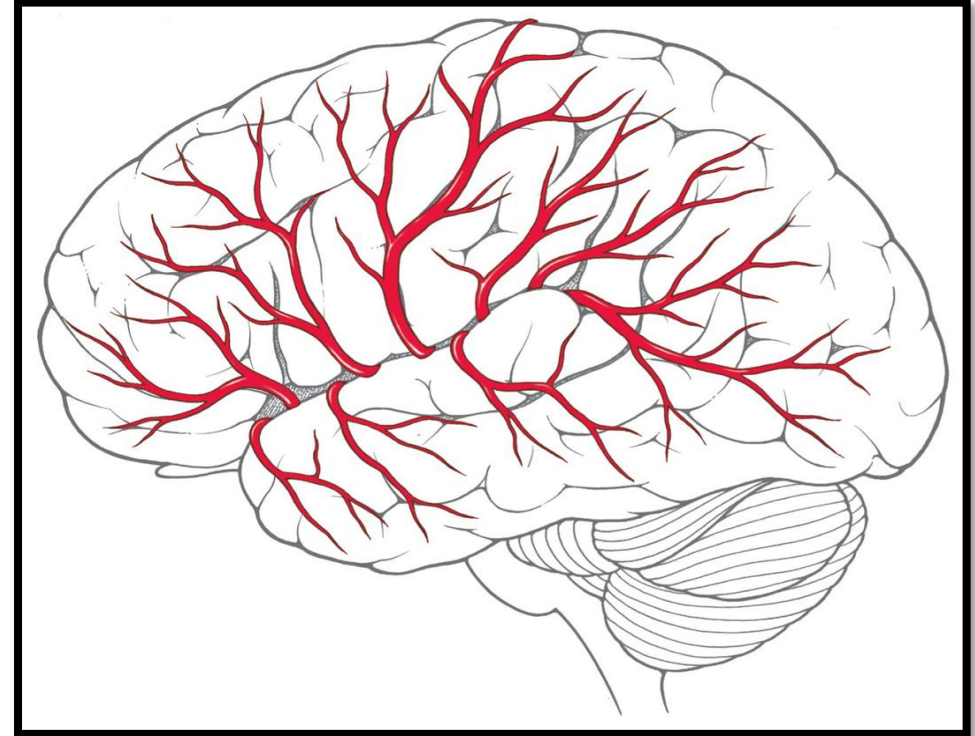
Arteries



Use the atlas and models to identify the arteries shown below.



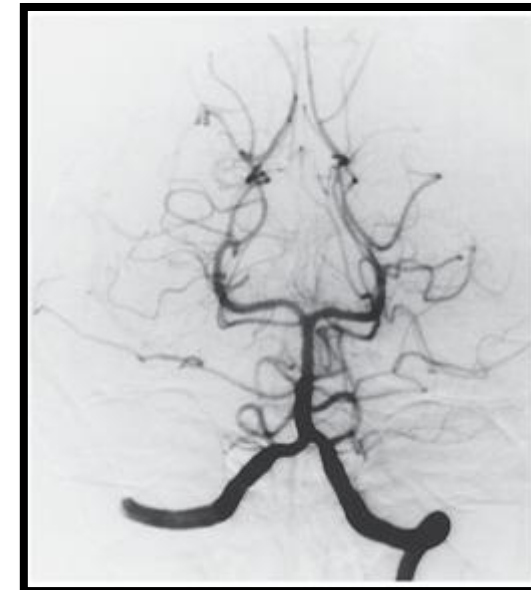
Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 10e, 2019 Fig 2-27



Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 10e, 2019 Fig 2-12

Angiogram

- A 57-year-old man comes to the emergency department because of intermittent left-side headache for the past 3 years and right upper limb numbness for 1 week. Angiography shows an aneurysm of the MCA. Identify this vessel on the angiogram shown and determine the reason why headache and motor symptoms are on opposite sides.
- Identify also, the Anterior cerebral, basilar, vertebral anterior inferior cerebellar, posterior inferior cerebellar and Internal carotid arteries on either of the angiograms.



Angiogram

Identify the numbered arteries from slide 13 on the angiograms below



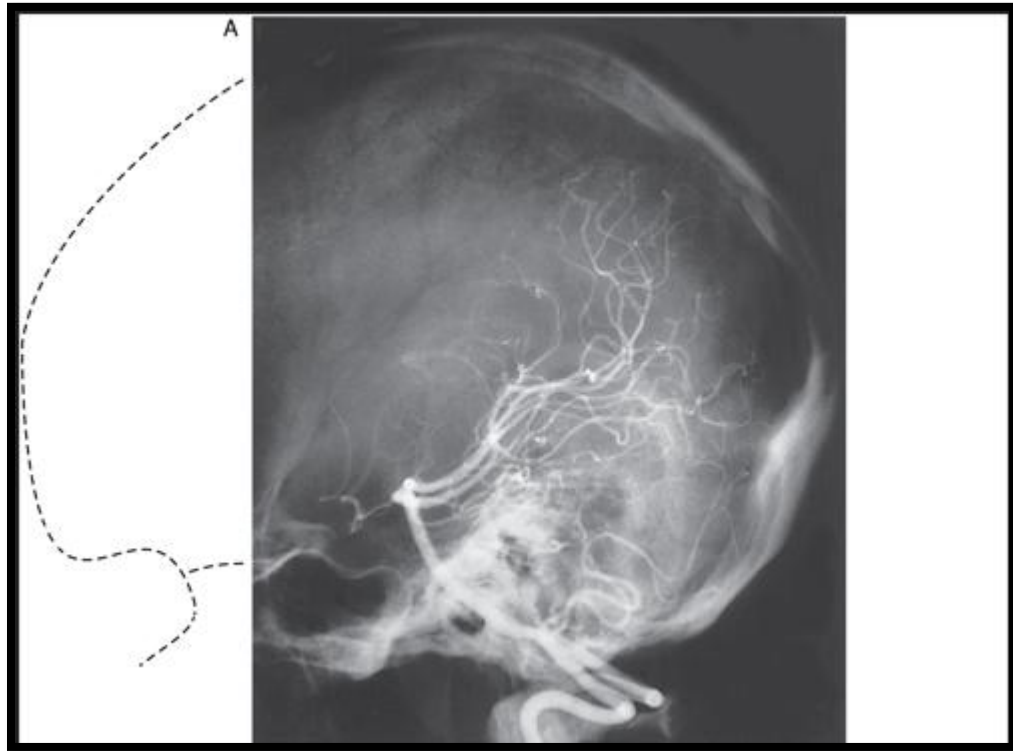
Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 10e, 2019 Fig 10-1a



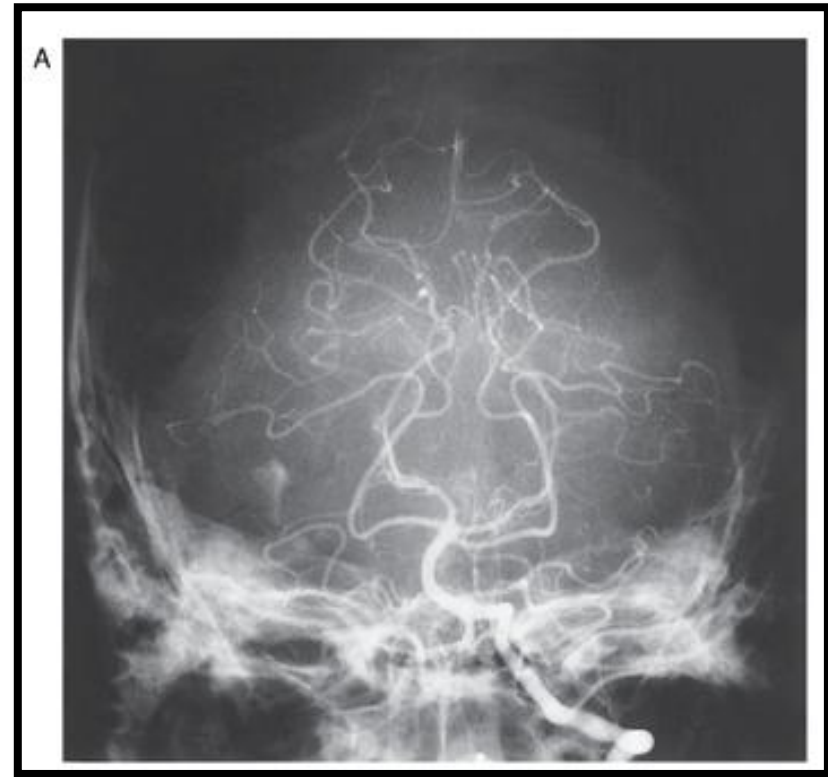
Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 10e, 2019 Fig 10-3a

Angiogram

Identify the numbered arteries from slide 13 on the angiograms below



Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 10e, 2019 Fig 10-7a



Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 10e, 2019 Fig 10-8a

Arteries

Analyze the lists below and match the artery with the area supplied and deficit that are likely to occur if there is a lesion to the artery.

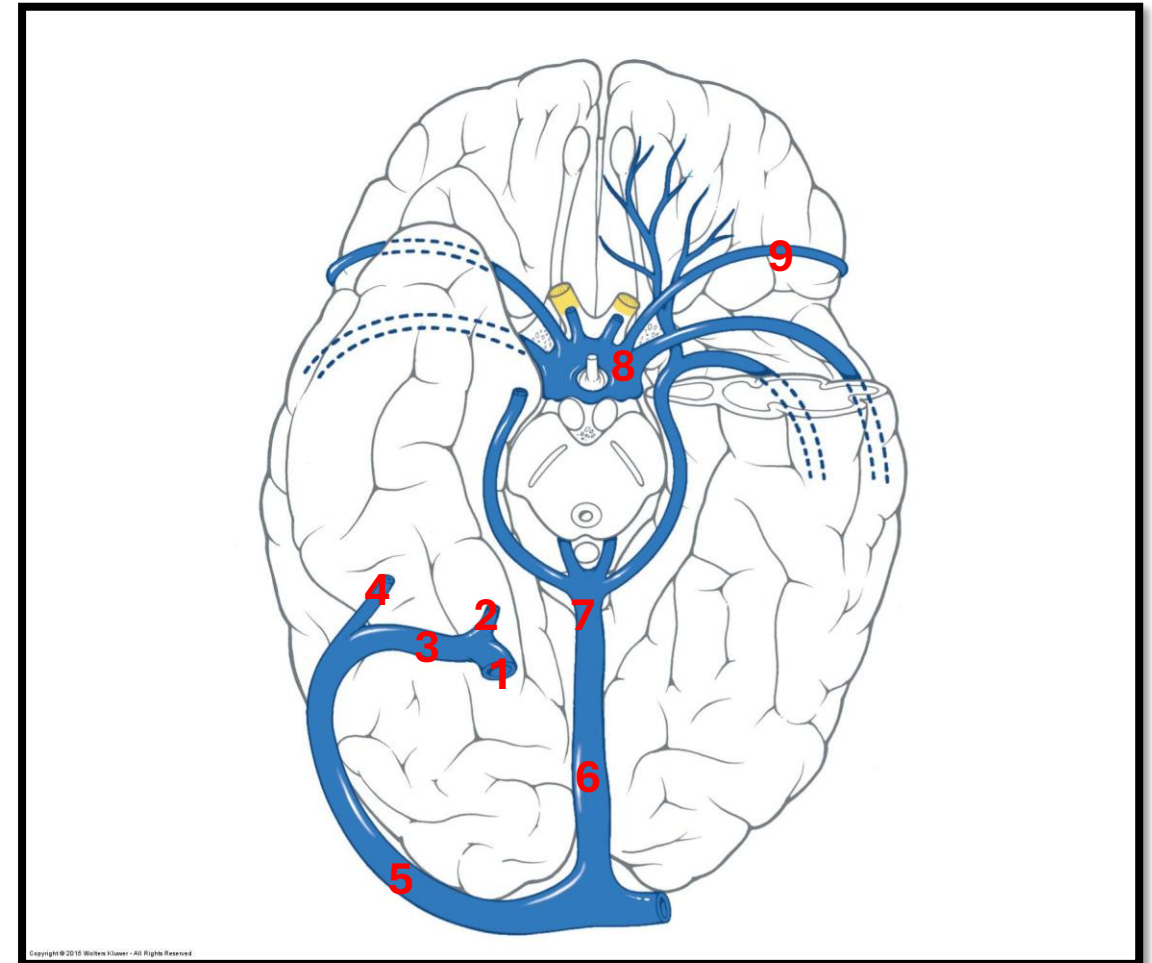
Artery	Area Supplied	Deficit
Anterior Cerebral Artery	Lateral frontal and parietal Lobes	Partial Loss of Vision
Middle Cerebral Artery	Paracentral Lobule	Motor and sensory loss of the of lower limbs
Posterior Cerebral Artery (PCA)	Medial Occipital Lobe	Motor and Sensory loss of the upper limbs
Posterior Inferior Cerebellar Artery	Cranial Nerve VII and VIII	Dysphagia
Anterior Inferior Cerebellar Artery	Dorsolateral Medulla	Facial Weakness and Vertigo

Veins and Sinuses



Use the atlas and models to identify the veins/sinuses shown below and the area they drain.

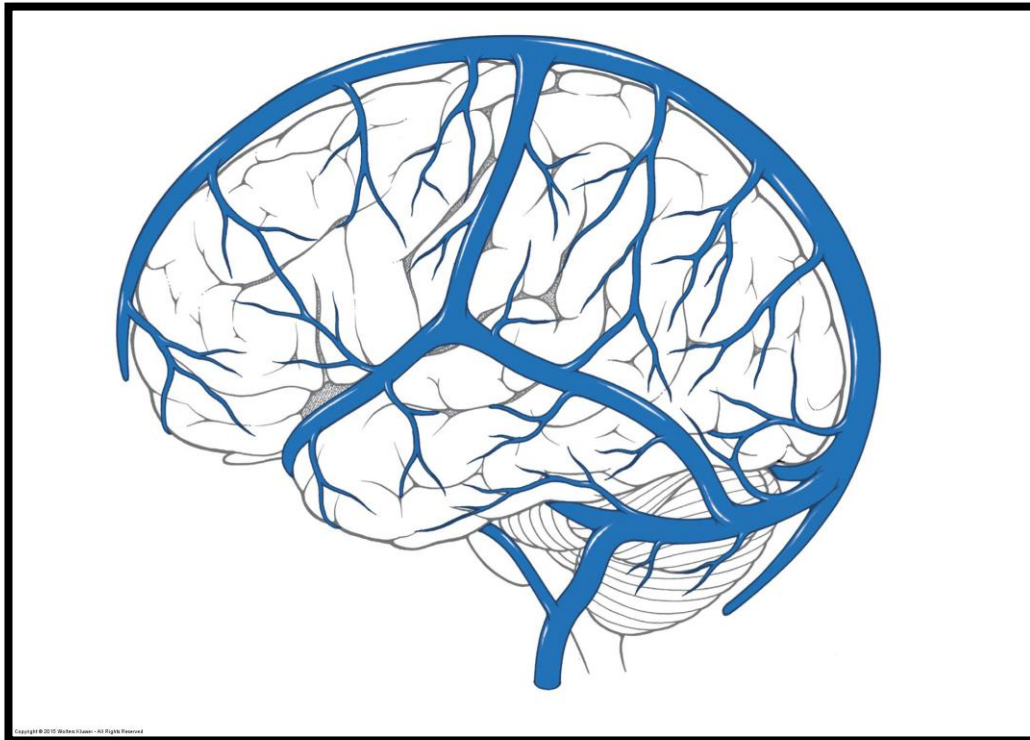
	Sinuses
1	
2	
3	
4	
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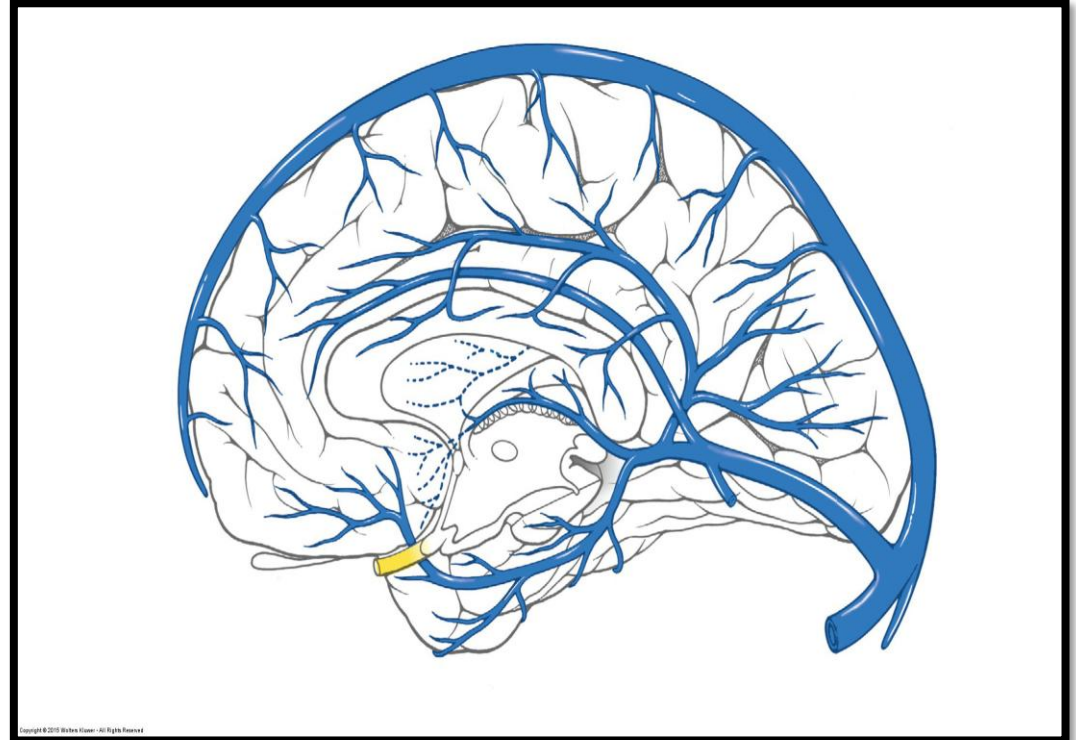
Veins and Sinuses



Use the atlas and models to identify the sinuses shown



Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 10e, 2019 Fig 2-13



Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 10e, 2019 Fig 2-28

Venogram

- A 27-year-old woman is brought to the ED because of severe headache and blurred vision for one week. Neurologic examination shows bilateral papilledema. Venography shows right transverse sinus and superior sagittal sinus thrombosis.
- Identify the transverse and superior sagittal sinuses on the venogram.



Neuroanatomy Atlas in Clinical Context: Structures, Sections, Systems, and Syndromes, 10e, 2019 Fig 10-2

Introduction to Neurological Cases and Clinical Vignettes

- Age and sex of a patient (Example: A 44-year-old woman)
- Comes to the physician because of ... (Example: a sudden onset of double vision)
- History, family history, medication
- Physical and neurologic examination
- Imaging and laboratory findings

Common Neurologic Diseases and their Characteristics (1 of 2)

Disease Category	Examples	Onset	Time Course
Vascular	Ischemic stroke (loss of blood supply) Hemorrhagic stroke (bleeding)	Typically, rapid onset	Acute (ischemic = worst at onset; hemorrhagic = getting worse)
Degenerative	Alzheimer's disease, Parkinson's disease	Slow onset	Chronic (gradually getting worse)
Traumatic	Injuries	Rapid onset	Acute
Infectious-inflammatory	Meningitis, Multiple sclerosis, Guillain-Barré syndrome	Variable	Acute or chronic

Common Neurologic Diseases and their Characteristics (2 of 2)

Disease Category	Examples	Onset	Time Course
Neoplastic	Tumors (glioblastomas, meningiomas)	Slow onset	Chronic (gradually getting worse)
Toxic-metabolic-nutritional	Vitamin deficiency, Poisoning	Variable	Variable
Genetic-hereditary	Huntington's disease	May be present at birth, or manifest later in life	Chronic

Note: The disease categories are not exclusive, there are overlaps. For example, Alzheimer's disease is degenerative, but may have a genetic component; Parkinson's disease is degenerative and may be caused by environmental toxins.

Signs and Symptoms - Terminology

- **Symptoms** describe the **subjective** experience of a disease, condition or injury by the patient (example: headache), and usually described by the patient in a clinical setting
- **Signs** describe the **objective** indication of a disease, condition or injury, usually observed or measured by a physician or another healthcare professional, using physical examination (increased blood pressure), neurologic examination (absence of deep tendon reflexes), lab results (elevated blood sugar), or imaging (aneurysm shown in an angiogram)
- **Cardinal signs** are signs specific to a certain disease (example: resting tremor, rigidity and slowness of movement are cardinal signs of Parkinson's disease)

Questions related to the Clinical Case

1. How is the condition of the patient in this video assessed?
2. What are the signs and symptoms of the disease?
3. Based on the onset and time course of the disease process, what is the most likely category of the disease?
4. Which structure of the nervous system is most likely impacted?
5. Point out the impacted structure(s) on the models provided.
6. Which structures related to the functions assessed are most likely still intact?
7. In case the lesion would have included additional structures in the vicinity, which functions would have been most likely impacted as well?